Module 15) Advance Python Programming

6. Class and Object (OOP Concepts)

- **Understanding the concepts of classes, objects, attributes, and methods in Python.**

1. Classes

- A **class** is a blueprint or template for creating objects.
- It defines a set of **attributes** (data) and **methods** (functions) that the created objects will have.
- Think of a class like a recipe, specifying what the object will contain and do.

2. Objects

- An **object** is an **instance** of a class.
- When a class is used to create a specific item, that item is an object.
- Each object can have different values for its attributes but shares the structure defined by the class.

3. Attributes

- **Attributes** are variables that belong to an object or class.
- They store data about the object's state or properties.
- Example: For a Car class, attributes might be color, model, and year.

4. Methods

- **Methods** are functions defined inside a class that describe the behaviors or actions of an object.
- They can manipulate object attributes or perform operations.
- Methods always have self as the first parameter, representing the object itself.

• Difference between local and global variables.

1. Global Variables

- Defined **outside of any function or block**.
- Accessible **from anywhere** in the code, including inside functions (unless shadowed).
- Have a **global scope** that lasts for the lifetime of the program.

Example:

```
    x = 10   Global variable

def func():
   print(x)

func()   Output: 10
```

2. Local Variables

- Defined **inside a function or block**.
- Accessible **only within that function** or block.

- Have a **local scope**, which means they cease to exist once the function finishes execution.

Example:

```
def func():
    y = 5   Local variable
    print(y)

func()   Output: 5
print(y)  Error
```